

Trading systems that **think** before **the trade**.

AI-native trading agency with a quant team. We have an open-source runtime (backtest-kit) where LLM reasoning and news sentiment analysis are used to find optimal entry points

0

look-ahead bias
AsyncLocalStorage

1.08

Sharpe ratio
TRX Jan 2026 case

100%

code parity
backtest = live

10+

LLM providers
OpenAI • Claude • Ollama



95% of trading bots use the same five strategies from tutorials

Crypto markets have entered second-order chaos.

Thousands of identical RSI / MACD / Bollinger bots execute the same signal in the same 100-millisecond window — they create the very noise they are trying to trade.

The only beneficiary is the exchange fee.

RSI · 85%

MACD · 78%

EMA cross · 72%

BB · 65%

Share of retail bots using each indicator. Combinations are also standard: RSI+MACD is the most popular.

-18% per bot / day

1,000 identical bots × 100 trades/day × -0.18% per round-trip.

Exchange earns +\$8,000/day.



Anatomy of self-destruction: one algorithm against itself

```
// Every 100ms on the exchange: 09:00.000
BTC drops to $87,300 09:00.050
1000 bots see: RSI < 30
→ "RSI in oversold zone"
→ "This is a buy signal" 09:00.100
All send BUY 09:00.150
Price spikes to $87,450 09:00.200
RSI instantly = 75
→ "RSI in overbought zone"
→ "This is a sell signal" 09:00.250
All send SELL 09:00.300
Price back to: $87,300
```

RESULT PER BOT

-0.18%

-0.1% spread – 0.08% fees per round-trip

RESULT FOR EXCHANGE

+\$8,000

0.08% × 1,000 bots × 100 cycles/day

PREDICTABLE STOP-LOSSES

90% of bots place SL at $ATR \times 1.5$. Market makers see the clusters and hunt them.



02 / METHODOLOGY

Four engineering solutions that **compound the edge**

Strategy alpha decays. The infrastructure beneath it does not. We invest in what makes every iteration verifiable, every signal reproducible, every run seamless.



Look-ahead bias is architecturally impossible

Every data request passes through the temporal context

`AsyncLocalStorage`. The strategy physically cannot access future data — the runtime will deny access.

```
// No timestamp parameters.  
// Context is propagated automatically.  
const c1h = await getCandles(s, "1h", 100);  
const c15m = await getCandles(s, "15m", 100);  
const c5m = await getCandles(s, "5m", 100);  
// → all three are synchronized  
// to the same backtest tick
```

`AsyncLocalStorage``Promise.all-safe`

Context is preserved across all async boundaries, including parallel requests

`Backtest``Paper``Live`

Same strategy code. The only difference is the source of `currentTime`

Multi-timeframe without errors

Synchronization of 1h / 15m / 5m data happens automatically through context



LLM searches news before making a decision

Eight specialized agents query Tavily and open sources for triggers — SEC actions, exchange hacks, presidential statements — over the past 4–12 hours.

Reasoning + Acting (ReAct) — the agent alternates between reasoning and acting. Between reasoning steps it searches the news feed.

ReAct loop

Tavily API

Zod validation

```
// Level 1 — Acute events
await search("BTC news ${date}");
await search("SEC DOJ action ${date}");
await search("exchange hack ${date}");

// Level 2 — Macro
await search("Fed surprise ${date}");

// Level 3 — Volume
await search("BTC volume spike ${date}");
```



Pine Script runs where TradingView is unavailable

PineTS executes .pine files locally on any data. Strategies built for BTC on Binance deploy unchanged on emerging market exchanges.

```
npm start --  
--pine ./math/feb_2026.pine  
--timeframe 15m  
--limit 500  
--when "2026-02-28T00:00:00Z"  
--jsonl // → dump for LLM
```

UZSE

Uzbekistan

live

MSE

Mongolia

live

DSE

Dhaka, Bangladesh

live

GSE

Ghana

live

SGBV

Algeria

beta

BSE

Botswana

beta

60+ indicators · Pine Script v5/v6 · JSONL-native output



AI workflow for liquidation cascades

Claude Code analyzes news, price correlation and formulates a hypothesis — no iterative brute-force. The strategy explains market mechanics rather than hunting random correlations.

- **Feb 5 — Flash crash (\$60K)**
V-bounce +15%, \$700M in liquidations
- **Feb 10 — Secondary cascade**
\$71K → \$67K, ETF outflows
- **Feb 23 — Tariff shock**
\$67.7K → \$62.5K, bounce

Strategy: LONG after liquidation spike ($ATR \times 2 + Volume \times 1.8$). Fixed exits: TP +1.5%, SL -0.8%.

+4.28%
Total PNL

83.3%
Win Rate

4.58
Profit Factor

0.00%
Max Drawdown

SHARPE RATIO
0.84



Inversion of the liquidity harvesting algorithm

A Telegram channel recommends SHORT with x25 leverage — exactly at the 4H candle low. OSINT analysis reveals an algorithmic pattern whose inversion yields consistent profit.

STEP	ACTION
01	Export all public signals with timestamps
02	Algorithm identified: TP multipliers are always $\times 1.52 \rightarrow \times 1.74 \rightarrow \times 1.47 \rightarrow \times 1.50$
03	Volume anomaly 15 min before each post — channel opens positions before publishing
04	Invert the position, filter by midpoint of the 4H range

+8.54%
Portfolio PnL

1.08
Sharpe Ratio

8/8
inverted signals

< 45 min
hold time



Three agent decisions during the Iran crisis

Real outputs of the ReAct pipeline — news search, contradiction assessment, one signal with written reasoning.

APRIL 5, 2026

US Ultimatum to Iran

Geopolitics creates uncertainty. RSI=44, volume low, Fear&Greed=12. Signals are contradictory.

WAIT

conf: 67%

APRIL 8, 2026

Two-week ceasefire with Iran

Rally to \$72K, \$425M in short liquidations, volume >2M BTC/hour. Sharp bullish impulse.

BUY

conf: 91%

APRIL 9, 2026

Quantum threats + miner selling

Break above \$71K, but strong bearish factors: hash rate falling, put options dominate (premium ~17%).

WAIT

conf: 73%



Open stack: from signal to execution

```
github.com/tripolskypetr/backtest-kit v6.0.0 · MIT backtest-kit/
```

```
├─ @backtest-kit/cli      scaffold·dump·backtest
├─ @backtest-kit/pinets   Pine Script runtime
├─ @backtest-kit/graph    reactive source/output
├─ @backtest-kit/ui       trade lifecycle UI
├─ @backtest-kit/ollama   10+ LLM providers
├─ agent-swarm-kit        ReAct + tool-calling
├─ functools-kit          Cache·str·AsyncLocal
└─ garch                  GARCH volatility
```

Self-validating runtime

Look-ahead bias, missing SL, incorrect R:R — all caught by Zod validators and AsyncLocalStorage before signal dispatch

Parallel execution

Hundreds of strategies run through a shared exchange handler, shared risk validator pool, shared queue. Monorepo architecture

JSONL-native logs

Every tick, every reasoning chain, every risk rejection — in JSON Lines format. Claude Code reads them directly



Five service lines

01	AI Strategy Development Collaborative strategy development: market structure analysis, hypothesis formulation, Pine Script, calendar-frame validation. Monthly recalibration.	Engagement
02	Discretionary Asset Management Strategy deployment under your mandate. JSONL audit of every trade, real-time dashboard. Min. \$250K AUM.	Managed
03	Quant Team Training Workshops: AsyncLocalStorage, ReAct agents, Pine Script → TypeScript, swarm-kit deployment. 2/5-day intensives.	Training
04	Emerging Market Integration Scraping, candle synthesis, data pipeline for 8 exchanges without TradingView: UZSE, MSE, DSE, GSE and others.	Infrastructure
05	backtest-kit Enterprise License Commercial license, priority support, custom risk validators, SLA deployment. Self-hosted or cloud.	Platform



10 research articles

- 01

Look-Ahead Bias

AsyncLocalStorage as temporal context
- 02

Second-Order Chaos

Why 1000 RSI bots lose 18%/day
- 03

AI Gets Hands for Trading

Claude Code + local Pine Script
- 04

Why the Price Drops in a Single Candle

Gamma convexity and liquidation cascades
- 05

AI Workflow for Cascades

Claude /loop, calendar-month strategies
- 06

News Sentiment AI Blueprint

ReAct pattern for trading signals
- 07

News Sentiment as a Signal

Vector search, 24-hour window
- 08

Liquidity Harvesting Machine

Telegram channel inversion, Sharpe 1.08
- 09

Pine Script on Exchanges Without TradingView

UZSE, MSE, DSE — 8 regional exchanges
- 10

Why Your Broker Froze Your Deposit

DCA, averaging and hidden drawdowns

 TheOne Trade

Ready strategy in 30 seconds.

```
npx -y @backtest-kit/cli --init
```

GITHUB

[tripolskypetr/backtest-kit](https://github.com/tripolskypetr/backtest-kit)

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DOCUMENTATION

backtest-kit.github.io